

Computational Physics Final Project Report

3D Projectile Motion with and without Air Resistance

Student YIYANG LIU

Course SCIE6063001 – Computational Physics

1. Introduction

Projectile motion is one of the most fundamental topics in classical mechanics. It describes the motion of an object launched into the air under the influence of gravity. In real-world situations, projectile trajectories are also affected by aerodynamic drag, making the motion more complex and difficult to solve analytically.

The objective of this project is to develop a Python-based computational model capable of simulating three-dimensional projectile motion under both ideal and non-ideal conditions. The project investigates the influence of air resistance and various physical parameters on the trajectory of a projectile while utilizing numerical techniques learned in Computational Physics.

2. Physical Background

The projectile is modeled as a spherical object moving in three-dimensional space.

The position vector is:

$$\mathbf{r} = (x, y, z)$$

The velocity vector is:

$$\mathbf{v} = (v_x, v_y, v_z)$$

2.1 Motion without Air Resistance

In the ideal case, only gravity acts on the projectile.

$$dv/dt = (0, 0, -g)$$

where:

$$g = 9.81 \text{ m/s}^2$$

The projectile follows a parabolic trajectory.

2.2 Motion with Air Resistance

The drag force is modeled using the quadratic drag equation:

$$F_d = -(1/2) \rho C_d A |v|v$$

where:

ρ = air density

C_d = drag coefficient

A = cross-sectional area

v = velocity vector

The drag force always acts opposite to the direction of motion and continuously reduces the projectile's kinetic energy.

3. Computational Method

Analytical solutions become increasingly difficult when air resistance is included. Therefore, numerical methods are required.

This project utilizes the Fourth-Order Runge-Kutta (RK4) method.

The RK4 algorithm estimates the next state of the system using four intermediate slope calculations:

k_1

k_2

k_3

k4

The weighted average of these values produces a highly accurate numerical approximation.

Advantages of RK4 include:

- High numerical accuracy
- Good stability
- Efficient computation
- Suitable for nonlinear differential equations

4. Project Implementation

The project was implemented entirely in Python using Jupyter Notebook.

Libraries used:

- NumPy
- Matplotlib
- IPython Display
- Jupyter Notebook

The simulation allows users to modify the following parameters:

Parameter	Description	
	-	--
Motion Type	Horizontal / Oblique	
Initial Height	Starting height	
Initial Speed	Launch speed	
Launch Angle	Vertical launch angle	
Azimuth Angle	Horizontal direction angle	
Mass	Projectile mass	
Radius	Projectile radius	
Drag Coefficient	Aerodynamic resistance	
Air Density	Environmental condition	
Time Step	RK4 numerical step	

5. Program Workflow

The simulation follows the workflow below:

Start



Input Parameters



Select Motion Type



Select Drag Model



Initialize Position and Velocity



RK4 Numerical Integration



Update State Variables



Ground Collision Detection



Generate Results



Visualization



End

6. Proposal Stage Summary

During the proposal stage, the topic "3D Projectile Motion with and without Air Resistance" was selected and approved.

The following items were presented:

- Physical process description
- Mathematical background
- Computational method selection
- Potential applications
- Expected visualization results

The project belongs to the Classical Mechanics domain.

Applications include:

- Ballistics
- Sports physics
- Aerospace engineering
- Motion prediction systems

7. Milestone Stage Summary

The project milestones were divided into several stages:

Task	Status
	--
Topic Selection	Completed
Mathematical Model	Completed
RK4 Solver Development	Completed
Air Resistance Implementation	Completed
Visualization Development	Completed
Testing and Validation	Completed
Presentation Preparation	Completed

All milestones were completed before the final presentation stage.

8. Results and Visualization

8.1 3D Projectile Trajectory

[Insert Screenshot of 3D Trajectory]

The simulation successfully generates a three-dimensional projectile trajectory.

The projectile follows a curved path due to the combined effects of gravity and air resistance.

8.2 Comparison between Air Resistance and No Air Resistance

[Insert Screenshot of Comparison Plot]

The comparison demonstrates that air resistance significantly affects projectile motion.

Observed effects:

- Reduced horizontal range
- Reduced maximum height
- Reduced flight time
- Faster kinetic energy loss

8.3 Animation of Projectile Motion

[Insert Screenshot of Animation]

A dynamic animation was created to visualize the motion of the projectile in real time.

The animation confirms the correctness of the numerical implementation and improves the interpretability of the simulation results.

9. Parameter Analysis

To investigate parameter sensitivity, several control-variable experiments were performed.

Initial Speed

Increasing the initial speed increases:

- Flight time
- Maximum height
- Horizontal range

because the projectile possesses greater initial kinetic energy.

Launch Angle

Increasing the launch angle increases the vertical component of velocity.

Approximately 45° provides the greatest range under ideal conditions.

Initial Height

Higher launch positions increase flight time and allow the projectile to travel farther before reaching the ground.

Drag Coefficient

Increasing the drag coefficient produces stronger aerodynamic resistance.

As a result:

- Range decreases
- Maximum height decreases
- Velocity decays more rapidly

10. Error Analysis

Numerical simulations introduce discretization errors because continuous differential equations are approximated using finite time steps.

In this project:

$dt = 0.01 \text{ s}$

was selected to balance accuracy and computational efficiency.

Potential sources of error include:

Numerical truncation error
Floating-point precision limitations
Simplified drag model assumptions

Despite these factors, RK4 provides highly accurate and stable results.

11. Boundary Case Analysis

Several boundary cases were considered.

Case 1: No Air Resistance

$C_d = 0$

The projectile follows the ideal trajectory predicted by classical mechanics.

Case 2: Large Drag Coefficient

$C_d \rightarrow 1$

Strong aerodynamic forces significantly reduce the projectile range.

Case 3: Vertical Launch

Launch Angle = 90°

The projectile travels almost entirely in the vertical direction.

Case 4: Horizontal Launch

Launch Angle = 0°

The projectile performs horizontal projectile motion.

These boundary cases confirm that the simulation behaves consistently with physical expectations.

12. Discussion

The simulation successfully demonstrates the importance of computational methods in solving

real-world physical problems.

The results show that air resistance has a substantial influence on projectile behavior and cannot be neglected in realistic applications.

The RK4 method proved to be accurate, efficient, and stable throughout the project.

13. Future Improvements

Possible extensions include:

- Wind modeling
- Variable air density
- Magnus effect due to projectile spin
- Advanced collision models
- Real-time interactive visualization

These improvements would increase the realism of the simulation.

14. Conclusion

A complete three-dimensional projectile motion simulator was successfully developed using Python and the RK4 numerical method.

The project accurately models both ideal and non-ideal projectile motion, provides numerical solutions to the governing equations, and visualizes the results through graphs and animation.

The results demonstrate the significant influence of air resistance on projectile trajectories and highlight the usefulness of computational physics techniques for solving complex physical systems.

The objectives of the project were successfully achieved.